

MTH 207 Exam 1 (Dr. Kim, Fall 2026)

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Rules

- The cover page should only have your name and email. Nothing else may be written on the cover page.
- No calculators.
- No notes, books, or other outside resources.
- No phones, smartwatches, earbuds, or any other device that communicates.
- Eyes on your own exam.
- Scratch paper is available upon request. You may only use the provided scratch paper, and all scratch paper (used and unused) must be submitted with your exam.

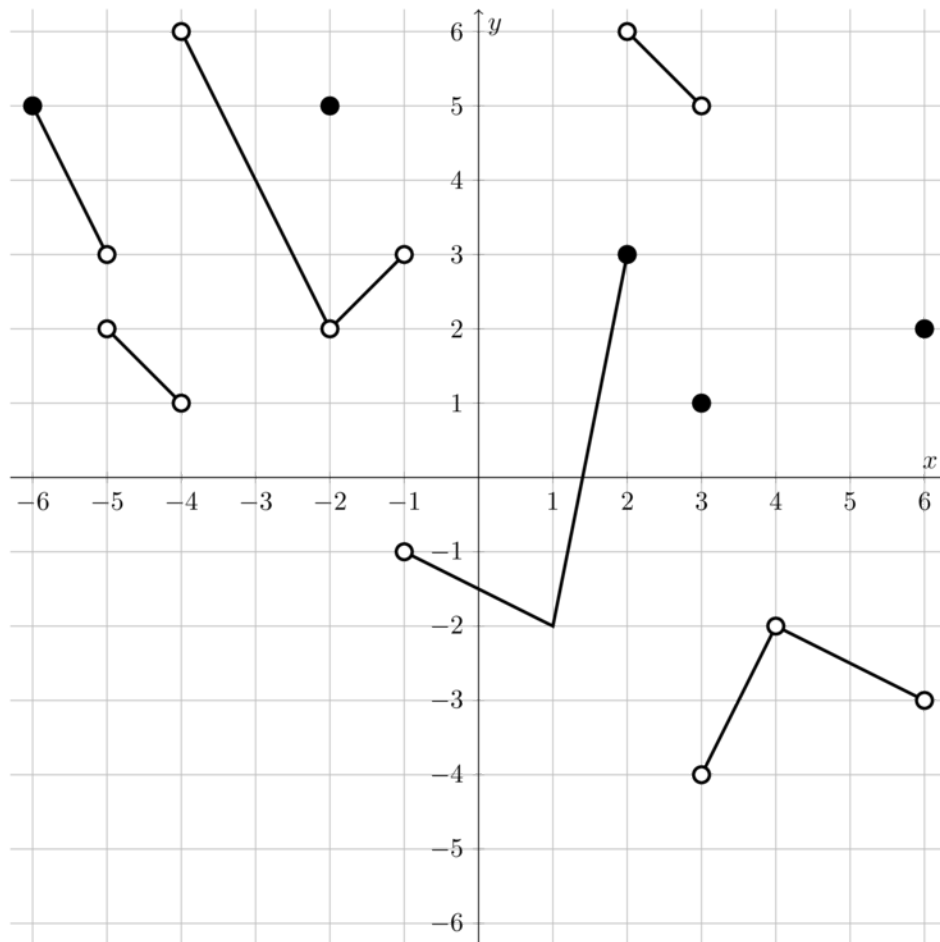
Expectations

- Write with proper notation.
- Show your work in order to earn points. University mathematics focuses on the **process** more than the **answer**. For example, stating “does not exist” without giving a reason will not earn points.
- *There is no guarantee that the practice exam has exactly all the same types of problems as the actual exam. (However, the practice exam is a good study resource for the actual exam.)*

The exam length is 55 minutes. There are 100 possible points.

1. The graph below is $y = f(x)$. If a requested value does not exist, explain why.

1



(3pts) (a) Find $\lim_{x \rightarrow -2} f(x) = 2$

2

3

4

5

6

(3pts) (b) Find $f(3) + f(6) = 1 + 2 = 3$

7

8

9

10

(3pts) (c) Find $\lim_{x \rightarrow 4} f(x) = -2$

11

12

13

14

(3pts) (d) Find $\lim_{x \rightarrow -4} f(x)$ dne

15

because $\lim_{x \rightarrow -4^-} f(x) \neq \lim_{x \rightarrow -4^+} f(x)$

16

17

18

19

- (8pts) 2. Given $f(x) = x^2$ find $f'(10)$ directly by a limit-based definition. (No credit will be given for applying derivative rules such as the Power Rule.)

$$f'(10) = \lim_{h \rightarrow 0} \frac{f(10+h) - f(10)}{h} = \lim_{h \rightarrow 0} \frac{(10+h)^2 - 10^2}{h} = \lim_{h \rightarrow 0} \frac{100 + 20h + h^2 - 100}{h}$$

$$= \lim_{h \rightarrow 0} \frac{20h + h^2}{h} = \lim_{h \rightarrow 0} \frac{h(20+h)}{h} = \lim_{h \rightarrow 0} (20+h) = 20$$

- (8pts) 3. Find m and c so that

$$f(x) = \begin{cases} 2x - m & \text{if } x < 6 \\ x - c & \text{if } x = 6 \\ 10 - x & \text{if } x > 6 \end{cases}$$

is continuous at $x = 6$, applying the definition.

① $f(6) = 6 - c$ ✓

② $\lim_{x \rightarrow 6^-} f(x) = \lim_{x \rightarrow 6^-} (2x - m) = 12 - m$

$\lim_{x \rightarrow 6^+} f(x) = \lim_{x \rightarrow 6^+} (10 - x) = 10 - 6 = 4$

$$\begin{aligned} 12 - m &= 4 \\ 8 &= m \end{aligned}$$

③ $6 - c = 4$
 $2 = c$

So $c = 2$ and $m = 8$

(8pts) 4. Find $\lim_{x \rightarrow 8} \frac{x-8}{x^2-64}$

$$\lim_{x \rightarrow 8} (x-8) = 0 \quad \lim_{x \rightarrow 8} (x^2-64) = 0$$

$$= \lim_{x \rightarrow 8} \frac{x-8}{(x+8)(x-8)}$$

$$= \lim_{x \rightarrow 8} \frac{1}{x+8}$$

$$\lim_{x \rightarrow 8} 1 = 1 \quad \lim_{x \rightarrow 8} (x+8) = 16$$

$$= \frac{\lim_{x \rightarrow 8} 1}{\lim_{x \rightarrow 8} (x+8)}$$

$$= \frac{1}{16}$$

(8pts) 5. Find $\lim_{u \rightarrow 3} \frac{u^2}{u+2}$

$$\lim_{u \rightarrow 3} u^2 = 9 \quad \lim_{u \rightarrow 3} (u+2) = 5$$

$$= \frac{\lim_{u \rightarrow 3} u^2}{\lim_{u \rightarrow 3} (u+2)}$$

$$= \frac{9}{5}$$

(8pts) 6. Find $\lim_{x \rightarrow 1} \frac{1}{(x-1)^2}$

$$\lim_{x \rightarrow 1} 1 = 1 \quad \lim_{x \rightarrow 1} (x-1)^2 = 0$$

x	y
0.9	$\frac{1}{(0.9-1)^2} = \frac{1}{(-0.1)^2} = \frac{1}{0.01} = 100$
0.99	10000

x	y
1.1	100
1.11	10000

$$\lim_{x \rightarrow 1^-} \frac{1}{(x-1)^2} = \infty$$

$$\lim_{x \rightarrow 1^+} \frac{1}{(x-1)^2} = \infty$$

$$\lim_{x \rightarrow 1} \frac{1}{(x-1)^2} = \infty$$

(8pts) 7. Find $\lim_{x \rightarrow 6} \frac{1}{x-6}$

$$\lim_{x \rightarrow 6^-} | = -1$$

$$\lim_{x \rightarrow 6^+} (x-6) = 0$$

x	y
5.9	$\frac{1}{5.9-6} = -10$
5.99	-100

x	y
6.1	10
6.01	100

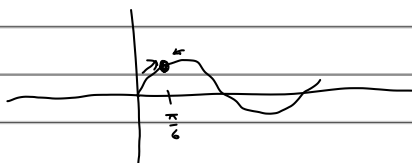
$$\lim_{x \rightarrow 6^-} f(x) = -\infty$$

$$\lim_{x \rightarrow 6^+} f(x) = \infty$$

$$\lim_{x \rightarrow 6} \frac{1}{x-6} \text{ dne}$$

(8pts) 8. Find $\lim_{x \rightarrow \frac{\pi}{6}} \sin(x)$

$$= \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$



Expectation Know trig values

If you feel like you keep forgetting,

we should talk about why $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$

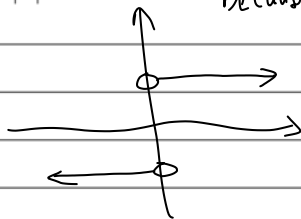
in office hours.

Parentheses required (order of operation)

(8pts) 9. Find $\lim_{x \rightarrow 9} \frac{x-9}{\sqrt{x}-3} \cdot \frac{\sqrt{x}+3}{\sqrt{x}+3} = \lim_{x \rightarrow 9} \frac{(x-9)(\sqrt{x}+3)}{x-9} = \lim_{x \rightarrow 9} (\sqrt{x}+3) = \sqrt{9}+3 = 3+3 = 6$

(8pts) 10. Find $\lim_{x \rightarrow 0} \frac{x}{|x|}$ dne because

$$\lim_{x \rightarrow 0^-} \frac{x}{|x|} \neq \lim_{x \rightarrow 0^+} \frac{x}{|x|}$$



$$\lim_{x \rightarrow 0^-} \frac{x}{|x|} = -1 \quad \text{and} \quad \lim_{x \rightarrow 0^+} \frac{x}{|x|} = 1$$

(8pts) 11. Find $\lim_{x \rightarrow 0} x^6 \sin(\frac{1}{x})$

Need $\leq$ not $<$

$$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$$

$$-x^6 \leq x^6 \sin\left(\frac{1}{x}\right) \leq x^6$$

$$\lim_{x \rightarrow 0} (-x^6) = 0$$

$$\lim_{x \rightarrow 0} x^6 = 0$$

$$\boxed{0 \leq x^6 \sin\left(\frac{1}{x}\right) \leq 0}$$

By the Squeeze Theorem, $\lim_{x \rightarrow 0} x^6 \sin(x) = 0$

(8pts) 12. Find $\lim_{x \rightarrow \infty} \frac{(4x^2 - 5)^{\frac{1}{2}}}{(8x^2 + 9)^{\frac{1}{2}}}$

$$= \lim_{x \rightarrow \infty} \frac{4 - \frac{5}{x^2}}{8 + \frac{9}{x^2}}$$

$$\lim_{x \rightarrow \infty} \left(8 + \frac{9}{x^2}\right) = 8 + 9 \cdot 0 = 8 \neq 0$$

$$= \frac{\lim_{x \rightarrow \infty} \left(4 - \frac{5}{x^2}\right)}{\lim_{x \rightarrow \infty} \left(8 + \frac{9}{x^2}\right)}$$

$$= \frac{4 - 5 \cdot 0}{8 + 9 \cdot 0} = \frac{4}{8} = \frac{1}{2}$$